

SPH4U Unit 2 Energy and Momentum

Chapter 5 Momentum and Collisions Lesson Package

Schoolwork	Useful Links/ Videos/ Handouts	Homework
Chapter 5 Videos		
	1. Collisions (a collection of videos)	
5.1 Momentum and Impulse		
1. Lesson 5.1 Presentation 2. Lesson 5.1 Lesson Package	Videos: 1. 5.1 Momentum and impulse 2. Impulse and Momentum 3. Introduction to Impulse & Momentum	1. 5.1 Check Your Understanding 2. Textbook pg. 227 #1-12 3. TIPERS pg183 – 185 B5-RT01, B5-RT02, B5-RT03, B5--RT04; pg. 187 B5-RT08; pg. 192-193 B5-RT13, B5-SCT14, B5-WWT15, B5-WWT16; pg. 194 B5-WWT17, B5-RT18
5.2 Conservation of Momentum in One Dimension		
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5.4 Head on Elastic Collisions		
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5.5 Collisions in Two Dimensions: Glancing Collisions		
1. Lesson 5.5 Presentation 2. Lesson 5.5 Lesson Package	Videos: 1. 5.5 Collisions in Two Dimensions: Glancing Collisions	1. Textbook pg. 253 #1-8 2. TIPERS pg. 201-202 B5- RT28, B5-QRT29
Chapter 5 Review		
1. Chapter 5 Self- Quiz pg. 263 (attempt all) 2. Chapter 5 Review pg. 264 #1-57 3. Communication Questions (lesson package)		1. Chapter 2 Self- Quiz pg. 263 (attempt all) 2. Chapter 2 Review pg. 264 #1-57 3. Chapter 5 - Check Your Knowledge – Definitions (lesson package) 4. Communication Questions (lesson package) 5. TIPERS

Lesson 5.1 Momentum and Impulse

Lesson Goal: The concepts of momentum and impulse will help you understand the science of collisions and answer the questions about collisions.

Success Criteria: At the end of this lesson, I will be able to:

- ✓ define momentum and impulse
- ✓ calculate the momentum and impulse of an object in motion
- ✓ describe the physical representation of the momentum and impulse of an object
- ✓ give examples where the direction of the impulse is different from the direction of the force that caused the change in momentum
- ✓ explain how the concepts of linear momentum and impulse are relevant to an understanding of motion in sports and in the design of improved sports gear.
- ✓ apply the concepts of momentum and impulse to analyse motor vehicle collisions and the motion of rockets.
- ✓ find the magnitude of an impulse by measuring the area under a force–time curve.

Linear momentum ($\vec{p}$) _____

Note: We will refer to linear momentum as momentum.



Note: The momentum $\vec{p}$ of an object is _____ to the object's velocity, so the momentum vector is _____ direction as the velocity. $\vec{p}$ is also _____ to the mass of the object.

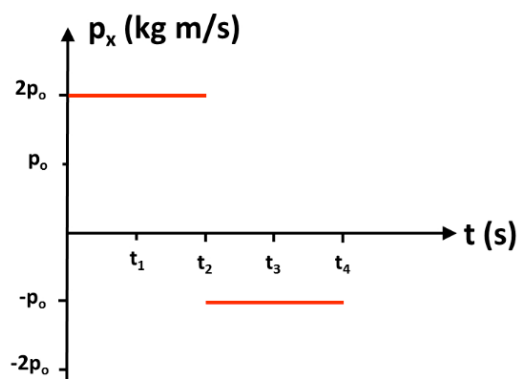
Example 1: The Vector Nature of Momentum

- a) Calculate the momentum of a 2.5 kg rabbit travelling with a velocity of 2.0 m/s [E].
- b) Calculate the momentum of a 5.0 kg groundhog travelling with a velocity of 1.0 m/s [S].
- c) Compare the momentum and kinetic energies of the rabbit and the groundhog.

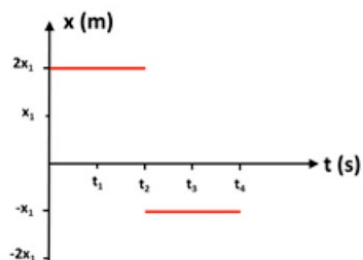
MC Question 1: A large semitrailer truck and a small car have equal momentum. How do their speeds compare?

- a) The truck has a much lower speed than the car.
- b) The truck has only a slightly lower speed than the car.
- c) Both have the same speed.
- d) The truck has only a slightly higher speed than the car.
- e) The truck has a much higher speed than the car.

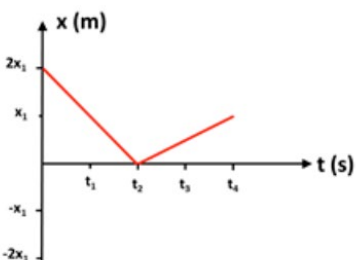
MC Question 2: The momentum versus time graph of a 170 g cue ball is shown in the figure.



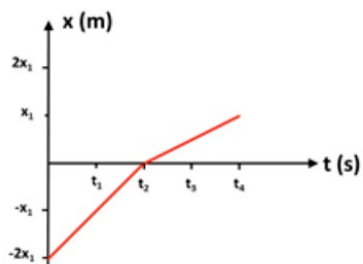
What is a possible corresponding position-time graph for the cue ball? (Hint: Momentum is directly proportional to velocity. How is velocity represented in a position-time graph?)



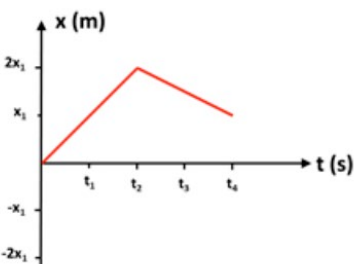
A



B



C



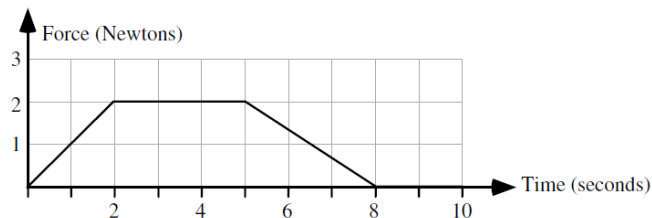
D

Impulse

Impulse and Force-Time Graphs

TIPERS B5-WWT19 (pg. 195): FORCE-TIME GRAPH II—IMPULSE APPLIED TO BOX

A 10-kg box, initially at rest, moves along a frictionless horizontal surface. A horizontal force to the right is applied to the box. The magnitude of the force changes as a function of time as shown.



A student calculates that the impulse applied by the force during the first 2 seconds is 4 N·s and that the impulse applied during the following 3 seconds is 6 N·s. What, if anything, is wrong with these calculations? If something is wrong, identify it and explain how to correct it. If these calculations are correct, explain why.

TIPERS B5-WWT20 (pg. 195): TWO SKATERS PUSHING OFF EACH OTHER—FORCE

Two skaters, a large girl and a small boy, are initially standing face-to-face but then push off each other. After they are no longer touching, the boy has more kinetic energy than the girl. A physics student who is watching makes the following contention about the forces that the boy and girl exerted on each other:

“Since the boy has more kinetic energy, he also has more momentum, so the girl had to have pushed harder on him than he pushed on her.”

What, if anything, is wrong with this contention? If something is wrong, identify all problems and explain how to correct them. If this contention is correct, explain why.

Question: Complete the sentences correctly.

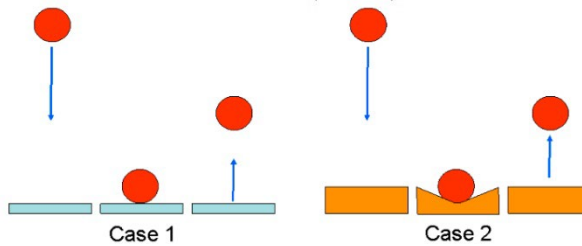
Stunt car A and stunt car B are identical cars with the same mass of 33.3 kg. They are both traveling at 55.1 m/s. Stunt car A crashes into a hardwood wall, whereas stunt car B crashes into a big pile of leaves. They both come to a complete stop after the impact.

Stunt car A experiences a _____ force over a _____ period of time.

Stunt car B experiences a _____ force over a _____ period of time.

Because of the force experienced by stunt car A, it will sustain _____ damage than stunt car B.

MC Question 3: Identical balls are dropped from the same initial height and bounce back to half the initial height. In Case 1, the ball bounces off a cement floor, and in Case 2, the ball bounces off a piece of stretchy rubber.



a) In which collision is the magnitude of the ball's change in momentum the greatest?

- i. Case 1
- ii. Case 2
- iii. Same in both

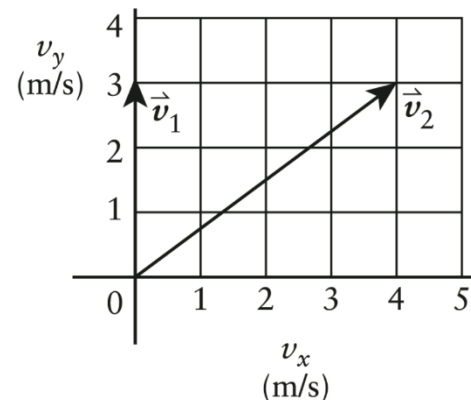
b) In which collision is the average force acting on the ball during the collision the biggest?

- i. Case 1
- ii. Case 2
- iii. Same in both

MC Question 4: An object of mass of 4.0 kg is initially moving with a velocity $\vec{v}_1$ as shown. A force is then exerted on it, which changes its velocity to $\vec{v}_2$, as shown.

a) The object's change in momentum has a magnitude of which of the following?

- i. 4.0 kg·m/s
- ii. 8.0 kg·m/s
- iii. 12.0 kg·m/s
- iv. 16.0 kg·m/s



b) What would be the change in kinetic energy of the object?

- i. 16 J
- ii. 18 J
- iii. 32 J
- iv. 64 J

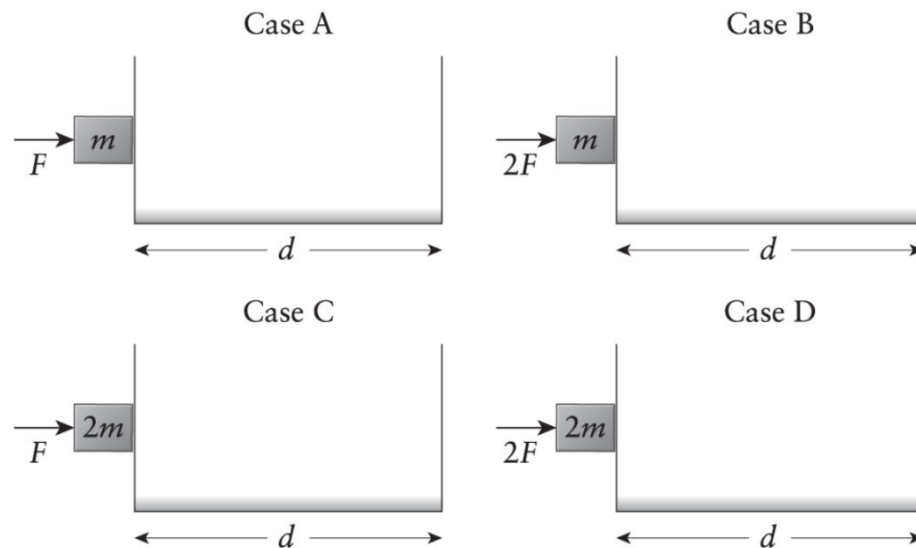
MC Question 5: The table below presents measurements of the net force exerted on a 0.50-kg object moving along a line during a 4.00-second time period.

$t(\text{s})$	0.00	0.50	1.00	1.50	2.00	2.50	3.00	3.50	4.00
$F(\text{N})$	2.01	2.00	2.01	3.00	4.02	5.00	4.02	2.99	2.00

Of the following, which would be the best estimate of the change in velocity for the object?

- a) 27 m/s
- b) 19 m/s
- c) 13 m/s
- d) 10 m/s

MC Question 6: A block is pushed a distance d across the floor in the four cases shown above. The block and the constant pushing force vary in value in each case as shown. The friction between the floor and the block is negligible in each case.



Which of the following is the correct ranking of the cases based on the momentum of the block after having been pushed a distance d ?

- a) $(B = D) > (C = A)$
- b) $(A = B) > (C = D)$
- c) $B > (A = D) > C$
- d) $D > (B = C) > A$

Example 2: Impulse as a Change in Momentum

A 0.160 kg puck is travelling at 5.0 m/s [N]. A slapshot produces a collision that lasts for 0.0020 s and gives the puck a velocity of 40.0 m/s [S].

- (a) Calculate the impulse imparted by the hockey stick.
- (b) Determine the average force applied by the stick to the puck.

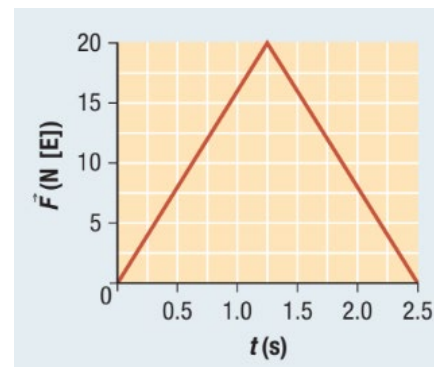
Example 3: Impulse as the Product of Force and Time

A volleyball player starts a serve by throwing the ball vertically upward. The 260 g volleyball comes to rest at its maximum height. The server then hits it and exerts an average horizontal force of magnitude 6.5 N on the ball.

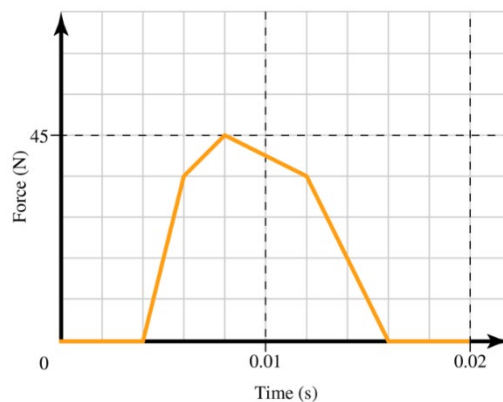
- (a) Determine the speed of the ball after the player hits it if the average force is exerted on the ball for 615 ms.
- (b) On the next serve, the volleyball player hits the ball with the same amount of horizontal force, but the time interval is 875 ms. Determine the speed of the ball.

Example 4: Impulse as Area under a Force–Time Curve

Two figure skaters approach each other in a straight line. They meet hand to hand and then push off in opposite directions. The increase and decrease of force are both linear, which produces a force–time curve that is in the shape of a triangle. The force–time curve for this interaction is shown in the figure below. Determine the impulse for this interaction.



Problem 1 (BFW): A 0.300 kg toy car moving with a speed of 0.820 m/s collides with a wall. The figure shows the force exerted on the car by the wall over the course of the collision. What is the magnitude of the velocity, or final speed, of the car after the collision?



Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding - Lesson 5.1

1. Why can a modern tennis racquet transfer a greater impulse than an older wooden-framed racquet?

5.2 Conservation of Momentum in One Dimension

Lesson Goal: You will examine what happens when two bodies interact such that momentum is exchanged.

Success Criteria: At the end of this lesson, I will be able to:

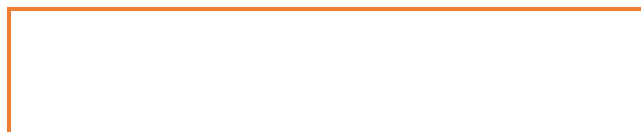
- ✓ differentiate between collision and explosion
- ✓ state the law of conservation of momentum both verbally and mathematically
- ✓ apply the law of conservation of momentum to calculate and predict the motion of objects after a collision or explosion
- ✓ explain the basic mechanics of rocket propulsion using the law of conservation of momentum

Collision _____

Explosion _____

The Law of Conservation of Momentum

Consider a system of two colliding objects as shown in the figure below.



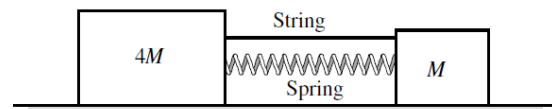
Law of Conservation of Momentum

Example 1: Collision Analysis

A hockey player of mass 97 kg skating with a velocity of 9.2 m/s [S] collides head-on with a defence player of mass 105 kg travelling with a velocity of 6.5 m/s [N]. An instant after impact, the two skate together in the same direction. Calculate the final velocity of the two hockey players.

TIPERs B5-CT21 (pg. 21): TWO BOXES ON A FRICTIONLESS SURFACE—MOMENTUM AND SPEED

Two boxes are tied together by a string and are sitting at rest on a frictionless surface. Between the two boxes is a massless compressed spring. The string tying the two boxes together is cut and the spring expands, pushing the boxes apart. The box on the left has four times the mass of the box on the right.

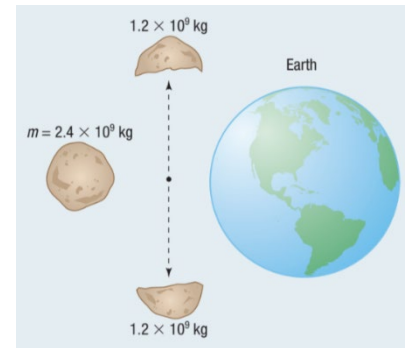


(a) After the string is cut and the boxes lose contact with the spring, will the magnitude of the momentum of the box on the left be (i) *greater than*, (ii) *less than*, or (iii) *equal to* the magnitude of the momentum of the box on the right? Explain your reasoning.

(b) At the instant (after the string is cut) that the boxes lose contact with the spring, will the speed of the box on the left be (i) *greater than*, (ii) *less than*, or (iii) *equal to* the speed of the box on the right? Explain your reasoning.

Example 2: Explosion Analysis

In a science fiction novel, a large asteroid is approaching Earth. Scientists decide to use explosive devices to blow the asteroid into two equal halves before impact with Earth. The asteroid has a mass of 2.4×10^9 kg. For each half to safely miss Earth, an explosion must cause each to travel a minimum of 8.0×10^6 m at a right angle away from Earth within 24 h (refer to the figure below). Assume this is a one-dimensional problem. The magnitude of the impulse applied to each fragment is the same, but the halves are directed in opposite directions.



- (a) Calculate the momentum of each part of the asteroid after the explosion.
- (b) Determine the impulse delivered to each part by the explosion.

MC Question 1: Two ice skaters, Karen and David, face each other while at rest, and then push against each other's hands. The mass of David is twice that of Karen. How do their speeds compare after they push off?

- a) Karen's speed is four times David's speed.
- b) Karen's speed is two times David's speed.
- c) Karen's speed is one-half of David's speed.
- d) Karen's speed is the same as David's speed.
- e) Karen's speed is one-fourth of David's speed.

MC Question 2: A soccer ball kicked due west during a practice collides with a rock that is initially at rest. Just after the collision, the ball is moving toward the southeast. Just after the collision, the rock must be moving

- a) toward the
- b) northeast.
- c) southeast.
- d) southwest.
- e) northwest.
- f) west.

Problem 1 (BFW): A stationary object explodes into two pieces: piece A has a mass M_A and piece B has a mass $M_B = 2M_A$.

a) Draw a before-and-after diagram to represent this explosion.

b) After the explosion, which piece has more momentum?

Piece A ____ Piece B ____ The magnitudes are the same. ____ Justify your choice.

c) After the explosion, which piece is moving faster?

Piece A ____ Piece B ____ They have the same speed. ____ Justify your choice.

d) After the explosion, which piece has more kinetic energy?

Piece A ____ Piece B ____ They have the kinetic energy. ____ Justify your choice.

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding – Lesson 5.2

1. Sometimes students are not sure which direction to select as the positive reference direction when solving questions involving linear momentum. What advice can you give to these students?
2. Use the concept of conservation of momentum to describe how the rocket's engine works in space where there is nothing to push on.
3. The difference between a collision and explosion.

5.3 Collisions

Lesson Goal: Explain the concept of work and describe what conditions must be met for work to be done on an object

Success Criteria: At the end of this lesson, I will be able to understand that:

- ✓ distinguish between elastic and inelastic collisions
- ✓ give examples of elastic and inelastic collisions
- ✓ explain why perfectly elastic and perfectly inelastic collisions do not happen
- ✓ explain the motion in Newton's cradle using the laws of conservation of momentum and energy

Elastic Collision _____

Conservation of kinetic energy _____

Inelastic Collision _____

Perfectly Elastic Collision _____



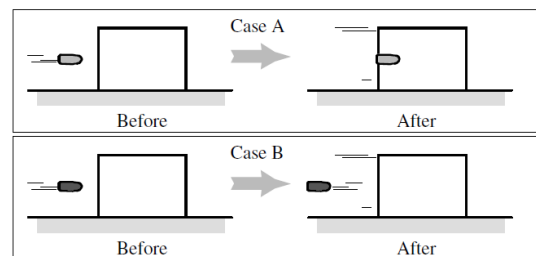
Perfectly Inelastic Collision _____



TIPERs B5-CT25 (pg. 198): BULLET STRIKES A WOODEN BLOCK—BLOCK AND BULLET SPEED AFTER IMPACT

In Case A, a metal bullet penetrates a wooden block. In Case B, a rubber bullet with the same initial speed and mass bounces off of an identical wooden block.

(a) Will the speed of the wooden block after the collision be (i) *greater* in Case A, (ii) *greater* in Case B, or (iii) *the same* in both cases? Explain your reasoning.



(b) In Case B, will the speed of the bullet after the collision be (i) *greater than*, (ii) *less than*, or (iii) *the same as* the speed of the bullet just before the collision? Explain your reasoning.

Example 1: Analyzing Perfectly Elastic Collisions

Suppose you have two balls with different masses involved in a perfectly elastic collision. Ball 1, with mass $m_1=0.26$ kg travelling at a velocity $v_1=1.3$ m/s [right], collides head-on with stationary ball 2, which has a mass of $m_2=0.15$ kg. Determine the final velocities of both balls after the collision.

Example 2: Applying the Conservation of Momentum to a One-Dimensional Perfectly Inelastic Collision

Two cars, a sports utility vehicle (SUV) of mass 2500 kg and a compact model of mass 1200 kg, are coasting at a constant velocity along a straight road. Their initial velocities are 40.0 m/s [W] and 10.0 m/s [W], respectively, so the SUV is catching up to the compact car. When they collide, the cars lock bumpers. Assume no external forces exist along the direction of travel, and the total momentum along x is conserved. Determine the velocity of the cars just after the collision.

Example 3: Applying Conservation of Momentum and Energy in a One-Dimensional Perfectly Inelastic Collision

A child with a mass of 22 kg runs at a horizontal velocity of 4.2 m/s [forward] and jumps onto a stationary rope swing of mass 2.6 kg. The child “sticks” on the rope swing and swings forward.

- (a) Determine the horizontal velocity of the child plus the swing just after impact.
- (b) How high do the child and swing rise?

Matching (BFW pg 407#7): Momentum and mechanical energy changes can be used to model the motion of objects within a system and also their external interactions. These conditions might be observed for the vector momentum and scalar mechanical energy.

- A) The momentum and mechanical energy changes are both zero.
- B) The momentum does not change and the change in mechanical energy is negative.
- C) The momentum does not change and the change in mechanical energy is positive.
- D) The momentum changes but mechanical energy does not.
- E) The momentum changes and the change in mechanical energy is positive.
- F) The momentum changes and the change in mechanical energy is negative.

Predict which of the conditions listed above would be observed in the following scenarios. If the scenario involves an interaction that causes the change, predict the change in terms of the value prior to the interaction subtracted from the value following the interaction. If a change in either momentum or energy is predicted to be negligible, include that expectation in your reasoning. Define your system in each situation.

- a) A golf ball initially at rest on the grass is struck by a golf club.
- b) Two hockey pucks slide over ice and collide.
- c) A bullet strikes a block of wood suspended by a rope.
- d) After two cars with the same mass approach each other at constant, but different, speeds, they collide and fuse, sliding together until they come to a stop.
- e) A balloon pops.
- f) A cart glides over a horizontal air track and bounces off the stationary rubber band stretched over the end of the track. A negligible friction force is exerted by the track on the cart and the collision of the cart with the rubber band is elastic.
- g) A cart glides repeatedly over a horizontal air track and bounces off stationary rubber bands stretched over the ends of the track until it finally comes to rest.
- h) A cart glides up an inclined air track, comes instantaneously to rest and then glides back down the track. The track exerts a negligible friction force on the cart.

MC Question 1: One object moves in the positive x direction with speed v . Another object moves in the opposite direction with the same speed but with one-fourth the mass of the first object. The two objects have a completely inelastic collision. What is the final x component of the velocity?

- a) 0
- b) $3v/5$
- c) $v/3$
- d) $v/2$
- e) v

MC Question 2: Two particles with the same mass of 11 kg are involved in a completely inelastic head-on collision. Before the collision, both particles have the same speed of 2 m/s. Which statement describes the velocities of the particles after the collision?

- a) Both particles have zero velocity.
- b) Each particle's speed is still 2 m/s, but each has reversed its direction.
- c) One of the particles continues to move at 2 m/s whereas the other particle is at rest.
- d) One of the particles continues at 2 m/s, and the other reverses direction and moves at 4 m/s.
- e) There is not enough information to determine the velocities after the collision.

MC Question 3: A hockey puck sliding on a frictionless surface strikes a box at rest. After the collision, the two objects stick together and move at some final speed. Which of the statements describes the change in momentum and energy of the puck during the collision?

- a) The puck loses some momentum in the collision but conserves its mechanical energy.
- b) The puck conserves its original momentum but loses all of its mechanical energy.
- c) The puck conserves its original momentum and loses some, but not all, of its mechanical energy.
- d) The puck loses some, but not all, of its original momentum and mechanical energy.
- e) The puck conserves its original momentum and mechanical energy.
- f) One cannot determine if the momentum or energy of the puck is conserved without knowing the final velocities and masses of the two objects.

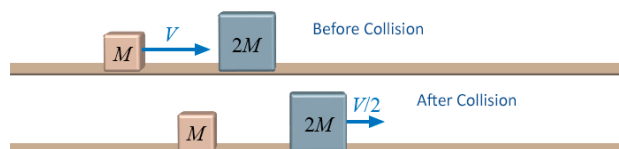
MC Question 4: Consider a completely elastic head-on collision between two particles that have the same mass and the same speed. What are the velocities of the particles after the collision?

- a) The magnitudes of the velocities are the same but the directions are reversed.
- b) Both velocities are zero.
- c) One of the particles continues with the same velocity, and the other comes to rest.
- d) One of the particles continues with the same velocity, and the other reverses direction at twice its previous speed.
- e) More information is required to determine the final velocities.

MC Question 5: A block having mass M slides on a frictionless surface and collides with a block of mass $2M$, which is initially at rest. After the collision, the small block is at rest, and the larger block is moving to the right with speed $V/2$.

Is this collision elastic?

- a) yes
- b) no
- c) depends on V



MC Question 6: Labeling two colliding objects as I and II, which of the following correctly describes collisions in general for unequal masses in which no external forces are exerted on the system?

☐	Change in momentum of each $\Delta \vec{p}_I = -\Delta \vec{p}_{II}$	Work done on each by the other $W_I \text{ in general } \neq W_{II}$
☐	Change in momentum of each $\Delta \vec{p}_I = \Delta \vec{p}_{II}$	Work done on each by the other $W_I \text{ in general } \neq W_{II}$
☐	Change in momentum of each $\Delta \vec{p}_I = -\Delta \vec{p}_{II}$	Work done on each by the other $W_I = -W_{II}$
☐	Change in momentum of each $\Delta \vec{p}_I = -\Delta \vec{p}_{II}$	Work done on each by the other $W_I = W_{II}$

Problem 1 (BFW -hw) A bullet with a mass $m_b=11.5$ g is fired into a block of wood at velocity $v_b=265$ m/s. The block is attached to a spring that has a spring constant k of 205 N/m. The block and bullet continue to move, compressing the spring by 35.0 cm before the whole system momentarily comes to a stop. Assuming that the surface on which the block is resting is frictionless, determine the mass of the wooden block.



Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

*

Check Your Understanding – Lesson 5.3

1. Explain what happens when two inelastic objects, say two snowballs, colliding head on.
2. Consider collisions in hockey, classify the specific collisions as elastic or inelastic, and explain why hockey is designed for those collisions to be elastic or inelastic.

5.4 Head-on Elastic Collisions

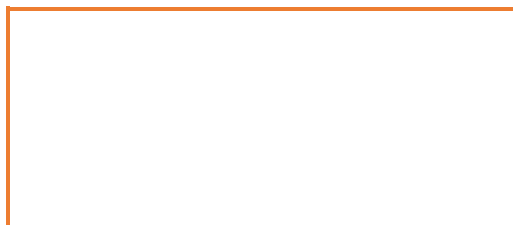
Lesson Goal: To examine in greater detail situations involving perfectly elastic collisions in one dimension and calculate the final velocities of two objects after an elastic collision in one dimension with equations for special cases.

Success Criteria: At the end of this lesson, I will be able to:

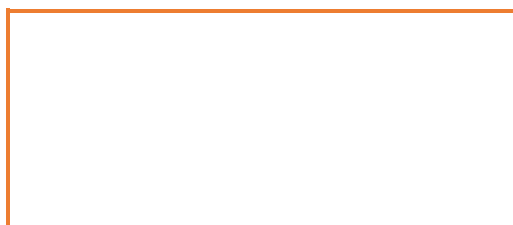
- ✓ apply the law of conservation of kinetic energy and law of conservation of momentum to derive the equations for final velocity of the two bodies in a head-on collision
- ✓ apply the derived equation and solve for the final velocity of objects in head-on collisions
- ✓ solve problems using these new equations for head-on elastic collisions in one dimension, special cases of collisions, such as objects of equal mass, produce some interesting results

Head-on Elastic Collision _____

The final velocity of the first object in terms of the masses and initial velocities of the two objects is given by



The final velocity of the second object in terms of the masses and initial velocities of the two objects.



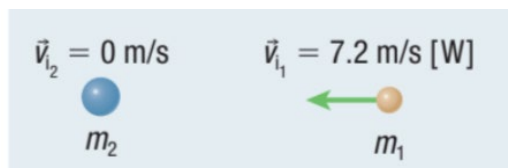
Note: The equation for $\vec{v}_{f2}$ is the same as the equation for $\vec{v}_{f1}$ if you interchange all the 1 and 2 subscripts. It is important to note that these equations hold true only for perfectly elastic collisions in one dimension.

In some cases, one of the objects is initially at rest. For instance, if v_2 is initially zero, the equations above simplify to



Example 1: Head-on Elastic Collision with One Object at Rest in One Dimension

Consider an elastic head-on collision between two balls of different masses, as shown in the figure below. The mass of ball 1 is 1.2 kg, and its velocity is 7.2 m/s [W]. The mass of ball 2 is 3.6 kg, and ball 2 is initially at rest. Determine the final velocity of each ball after the collision.



Example 2: Head-on Elastic Collision with Both Objects Moving in One Dimension

In a bumper car ride, bumper car 1 has a total mass of 350 kg and is initially moving at 4.0 m/s [E]. In a head-on completely elastic collision, bumper car 1 hits bumper car 2. The total mass of bumper car 2 is 250 kg, and it is moving at 2.0 m/s [W]. Calculate the final velocity of each bumper car immediately after the collision.

TIPERs B5-SCT24 (pg. 197): TWO MOVING CARTS—RESULT OF COLLISION

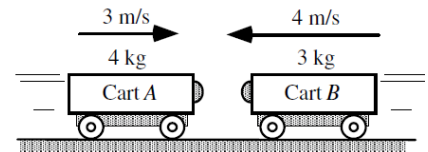
Carts A and B are shown just before they collide. Four students discussing this situation make the following contentions:

Alma: “After the collision, the carts will stick together and move off to the left. Cart B has more speed, and its speed is going to determine which cart dominates in the collision.”

Baxter: “I think they’ll stick together and move off to the right because Cart A is heavier. It’s like when a heavy truck hits a car: The truck is going to win no matter which one’s going fastest, just because it’s heavier.”

Callie: “I think the speed and the mass compensate, and the carts are going to be at rest after the collision.”

Dante: “The carts must have the same momentum after the collision as before the collision, and the only way this is going to happen is if they keep the same speeds. All the collision does is change their directions, so that Cart A will be moving to the left at 3 m/s and Cart B will be moving to the right at 4 m/s.”



With which, if any, of these students do you agree? Alma ____ Baxter ____ Callie ____ Dante ____ None of them ____

Explain your reasoning.

Special Cases

Using these new equations for head-on elastic collisions in one dimension, special cases of collisions, such as objects of equal mass, produce some interesting results.

CASE 1: OBJECTS HAVE THE SAME MASS

CASE 2: A LIGHTER OBJECT COLLIDING WITH A MUCH HEAVIER, STATIONARY OBJECT

Conservation of Mechanical Energy

What do you suppose happens to the conservation of total mechanical energy during elastic collisions?

One of the two gliders in Figure 3 has been fitted with a spring bumper. When the two gliders collide head-on in an elastic collision, the bounce is not immediate. If you viewed the collision in slow motion, you would see the bumper compress initially and then spring back to its original shape. During the compression, some of the kinetic energy of the moving gliders is converted into elastic potential energy. This potential energy is converted back into kinetic energy during the rebound.

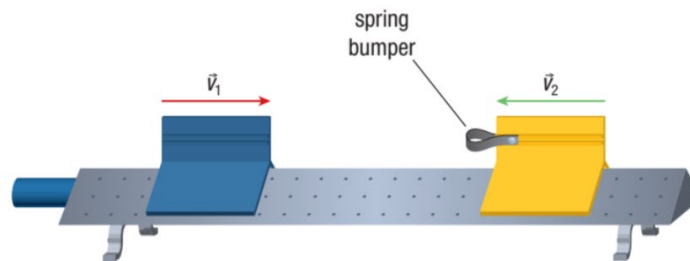


Figure 3 When the two gliders collide, the duration of the collision is greater than it would be without the spring bumper on one of the gliders.

If the compression of the spring bumper during the collision is x , then the law of conservation of energy states:

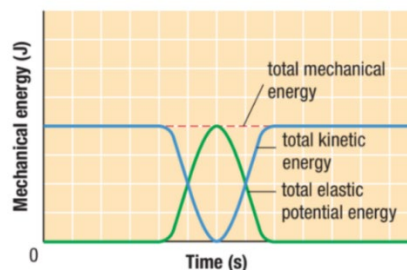


Figure 4 In this graph of total mechanical energy versus time, you can see how the total mechanical energy, the total kinetic energy, and the total elastic potential energy relate to each other throughout the collision.

To determine the maximum compression of the spring during the collision, use the fact that when the two gliders collide, they have the same velocity at that point. If they did not have the same velocity at maximum compression, then one would be catching up to the other or pulling away from the other. Therefore, **at maximum compression (closest approach), the two objects must have the same velocity, v_f .**

Example 3: Two-Cart Spring System

Dynamics cart 1 has a mass of 1.8 kg and is moving with a velocity of 4.0 m/s [right] along a frictionless track. Dynamics cart 2 has a mass of 2.2 kg and is moving at 6.0 m/s [left]. The carts collide in a head-on elastic collision cushioned by a spring with spring constant $k=8.0 \times 10^4 \text{ N/m}$ (refer to the figure below).

- (a) Determine the compression of the spring, in centimetres, during the collision when cart 2 is moving at 4.0 m/s [left].
- (b) Calculate the maximum compression of the spring, in centimetres.

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

5.5 Collisions in Two Dimensions: Glancing Collisions

Lesson Goal: Examine collisions in two dimensions.

Success Criteria: At the end of this lesson, I will be able to:

- ✓ express velocity and momentum as sums of their vertical and horizontal components
- ✓ calculate the sum of two or more velocities or momentums by summing their respective components
- ✓ apply the laws of conservation of momentum and conservation of kinetic energy for collisions in 2D
- ✓ apply the conservation of momentum for elastic and inelastic collisions.
- ✓ apply the conservation of kinetic energy in elastic collisions.
- ✓ solve 2D collision problems by independently analyzing the x-components and y-components.

Components of Momentum

Glancing Collision

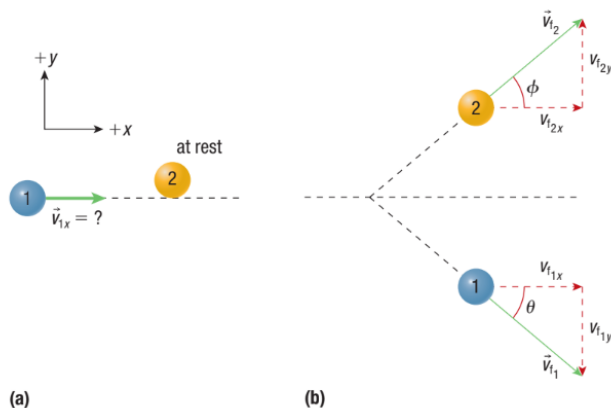
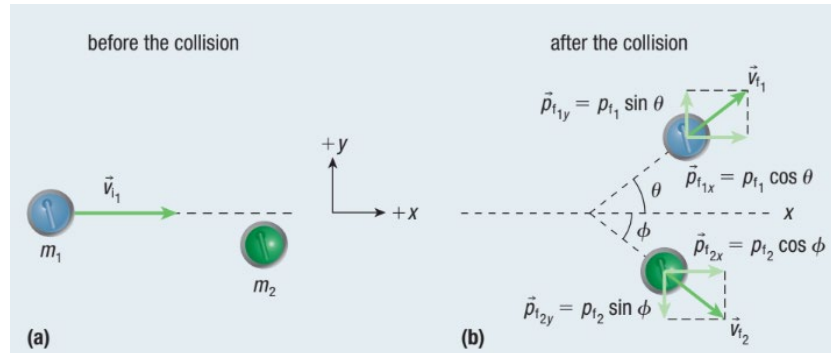


Figure 2 A cue ball striking another ball at an angle causes both balls to change direction.

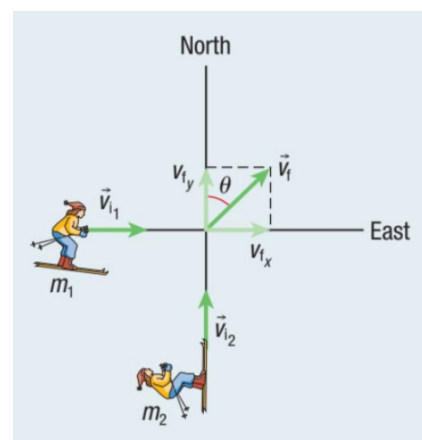
Example 1: Analysis of a Glancing Collision

In a game of curling, a collision occurs between two stones of equal mass. The object stone is initially at rest. After the collision, the stone that is thrown has a speed of 0.56 m/s in some direction, represented by θ in Figure 3. The object stone acquires a velocity $v_{f2} = 0.42$ m/s at an angle of $\phi = 30.0^\circ$ from the original direction of motion of the thrown stone. Determine the initial velocity of the thrown stone.



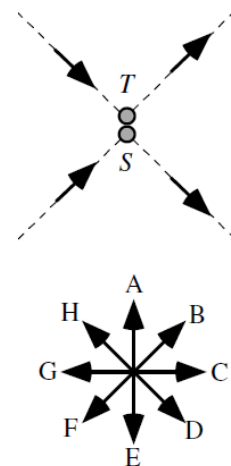
Example 2: Inelastic Glancing Collisions

Two cross-country skiers are skiing to a crossing of horizontal trails in the woods as shown in the figure on the right. Skier 1 is travelling east and has a mass of 84 kg. Skier 2 is travelling north and has a mass of 72 kg. Both skiers are travelling with an initial speed of 5.1 m/s. One of the skiers forgets to look, resulting in a right-angle collision with the skis locked together after the collision. Calculate the final velocity of the two skiers.



TIPERs B5-QRT27: COLLIDING STEEL BALLS—MOMENTUM AND IMPULSE DIRECTION

Two identical steel balls, *S* and *T*, are shown at the instant that they collide. The paths and velocities of the two balls before and after the collision are indicated by the dashed lines and arrows. The speeds of the balls are same before and after the collision. For the questions below, use the directions indicated by the arrows in the direction rosette, or use **J** for no direction, **K** for into the page, or **L** for out of the page.



(a) Which letter best represents the direction of the initial momentum of ball *T*? Explain your reasoning.

(b) Which letter best represents the direction of the final momentum of ball *T*? Explain your reasoning.

(c) Which letter best represents the direction of the change in momentum for ball *T*? Explain your reasoning.

(d) Which letter best represents the direction of the change in momentum for ball *S*? Explain your reasoning.

(e) Which letter best represents the direction of the impulse on ball *T*? Explain your reasoning.

(f) Which letter best represents the direction of the impulse on ball *S*? Explain your reasoning.

Chapter 5 Additional Problems

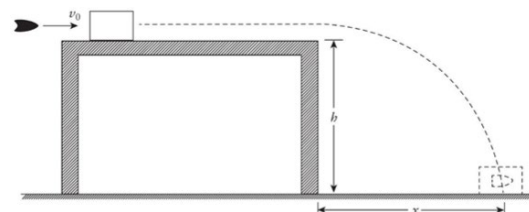
Lesson Goal: Apply the knowledge of the basic principles to problem-solving

Success Criteria: At the end of this lesson, I will be able to:

- ✓ Apply the knowledge to more challenging problem-solving questions

Problem 1 (BFW): A bullet of mass 0.016 kg is moving horizontally with speed $v_0 = 282\text{ m/s}$ when it hits a block of mass 1.2 kg that is at rest on a table, as shown. There is negligible friction between the block and the table. The surface of the table is 1.05 m above the floor. After the impact, the bullet and the block slide off the table and hit the floor a distance x from the edge of the table.

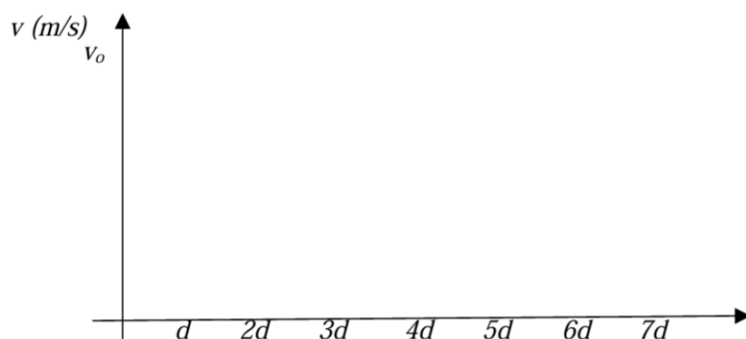
- Calculate the speed of the block (with the bullet in it) as it leaves the table.
- Calculate the mechanical energy dissipated in the collision.
- Suppose that the bullet passes through the block instead of remaining in it. State whether the distance x for the block would now be greater than, less than, or the same as when the bullet stayed in the block. Justify your answer.



Problem 2 (BFW) A cart, with a mass m , is moving along a track, with negligible friction, at a speed v_0 . Students drop sugar packets straight down onto the cart at regular distances, d . The mass of each sugar packet is $m/10$, and 5 packets are dropped.

a) i. How will the motion of the cart change as the packets are dropped? Support your answer with physics principles.

ii. On the axes below, sketch the velocity of the cart vs. the distance when the packets are dropped onto the cart and the time when no more packets are being dropped. Clearly label the graph.



b) Derive an expression for the final velocity of the cart after 5 sugar packets have been dropped individually into the cart. Give your answer in terms of m , v_0 , d , and fundamental constants.

In a second experiment, the same cart, moving with a speed of v_0 , moves along the track, but this time two packets are dropped at each distance interval of d .

c) i. How would the motion of the cart change for the second experiment, compared to the first experiment?

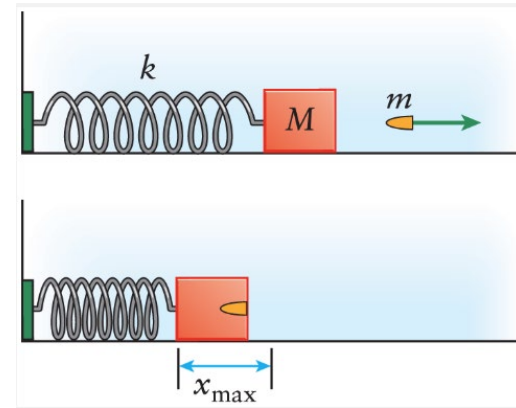
ii. On the axes above, sketch the motion of the second cart for the same distance intervals as in part (a). Clearly label the graph.

Problem 3 (BFW pg 431#21): In a ballistic pendulum experiment, a small marble is fired into a cup attached to the end of a pendulum. If the mass of the marble is 0.0075 kg and the mass of the pendulum is 0.25 kg, how high will the pendulum swing if the marble has an initial speed of 6 m/s? Assume that the mass of the pendulum is concentrated at its end and the velocity of the marble as it enters the cup is purely horizontal.

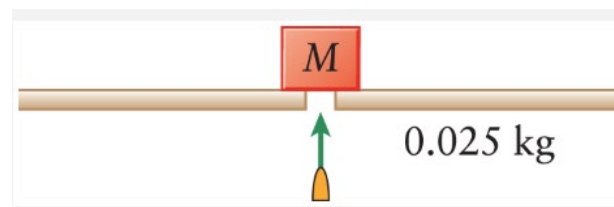
Problem 4 (BFW pg 431#24) A block of mass M , initially at rest, is attached to an ideal spring of constant k , which is attached to a wall. A bullet of mass m is fired into the block and the block moves, compressing the spring a distance $x_{\max}$ before coming to rest. Assume that friction between the block and the floor is negligible.

a) Derive an expression for the speed of the bullet just before it strikes the block, in terms of the given quantities and physical constants.

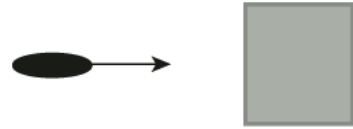
b) If the bullet is instead passed through the block, would the compression of the spring $x_{\max 2}$ be greater than, less than, or the same as for when the bullet remains embedded? Justify your answer.



Problem 5 (BFW pg 431#25): A bullet of mass 0.025 kg is fired at a speed of 250 m/s from below into a block of unknown mass, which is initially at rest on a horizontal surface. The block with the bullet embedded in it rises to a maximum height of 1.3 m before falling back downward. Calculate the mass of the block.



Problem 6 (BFW pg 434#1): A wooden block of mass m_2 is initially at rest on a rough table. The teacher gives the student two projectiles that can be fired toward the block, a metal dart and a hard rubber bullet, both of the same mass m_1 . The metal dart will embed in the block, and the block will then slide across the table and come to rest after moving a distance D_M . The rubber bullet will bounce off the block, and the block will then slide across the table and come to rest after moving a distance D_R .



- If the velocities of both the dart and the bullet are the same, v_1 , when they strike the block, sketch graphs of the velocity versus time of the block for the two cases on a sheet of graph paper.
- Assuming $m_1 = m_2 = M$, derive expressions for D_M and D_R in terms of v_1 and any necessary constants.
- Do your graphs in part (a) support your answer in part (b)?

Problem 7 (BFW pg 434#2): Two identical carts of mass m are set up to collide on a track in a physics classroom. Assume you may neglect friction between the carts and the track. Motion detectors are set up to measure the velocity of each cart both before and after the collision. The bumpers on the carts can be adjusted to change the elasticity of the collision.

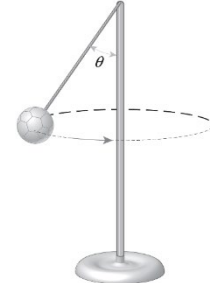
Three trials at different elasticity settings for the same initial velocities of magnitude v result in three different final conditions.

- In trial A, the two carts come to rest as a result of the collision.
- In trial B, the two carts bounce off each other and exchange velocities.
- In trial C, the two carts bounce off each other and move with speeds that are $2/3$ the magnitude of the speeds they had before the collision.

a) For each trial, identify the type of collision and describe the center-of-mass velocity before, during, and after the collision. Also describe any energy changes during the interaction, clearly defining your system.

b) For each trial, calculate the change in kinetic energy of the two carts for the collision.

c) Do your calculations in (b) support your claims in (a)? Briefly justify your answer.



Problem 8 (BFW pg434#3). Two students are playing a game called tetherball, where a ball of mass M_b is connected (tethered) by a rope to the top of a vertical pole, so that it can move around the pole, as shown in the figure.

As the tethered ball passes directly in front of the pole, such that it is moving to the right with speed v_0 , it collides head-on with, and sticks to, a small lump of clay of mass M_c . The clay is thrown at it softly by a student, such that its velocity upon impact can be approximated as 0. The rope breaks at the same instant the clay collides with the tethered ball.

- a) On a sheet of graph paper, sketch a graph of the speed of the tetherball from just before the collision until the tetherball–clay system strikes the ground.
- b) The tetherball is at a height H when the rope breaks. In terms of the given and fundamental constants:
 - (i) Derive an expression for the displacement of the center of mass of the system as a function of time from immediately after the collision until just before the system strikes the ground.
 - (ii) Derive an expression for the displacement of the center of mass of the system as the system strikes the ground.
- c) Relate your expression in (b) to your graph in (a).

Lesson Reflection

List the problems you knew how to do:

*

List the problems you need to go over again:

*

Chapter 5 - Check Your Knowledge – Definitions

Term	Definition
___ glancing collision	1. an impact in which two objects approach each other from opposite directions; momentum and kinetic energy are conserved after the collision
___ inelastic collision	2. a quantity that describes the motion of an object travelling in a straight line as the product of its mass and velocity
___ perfectly elastic collision	3. an ideal collision in which two objects stick together perfectly so they have the same final velocity; in this situation, momentum is perfectly conserved, but kinetic energy is not conserved
___ conservation of kinetic energy	4. an ideal collision in which external forces are minimized to the point where momentum and kinetic energy are perfectly conserved
___ collision	5. the product of force and time that acts on an object to produce a change in momentum
___ law of conservation of momentum	6. a collision in which momentum and kinetic energy are conserved
___ explosion	7. a collision in which the first object, after the impact with the second object, travels at an angle to the direction it was originally travelling
___ linear momentum	8. the impact of one body with another
___ impulse	9. a collision in which momentum is conserved, but kinetic energy is not conserved
___ elastic collision	10. When two objects collide in an isolated system, the collision does not change the total momentum of the two objects. Whatever momentum is lost by one object in the collision is gained by the other. The total momentum of the system is conserved.
___ perfectly inelastic collision	11. the total kinetic energy of two objects before a collision is equal to the total kinetic energy of the two objects after the collision
___ head-on elastic collision	12. a situation in which a single object or group of objects breaks apart

Chapter 5 –Communication Questions

1. Momentum and kinetic energy are both calculated from the mass and velocity of an object. If mass is conserved during a collision, explain how momentum may be conserved but not kinetic energy.
2. The collision examples in chapter 5 only considered idealized, isolated systems. What are the problems with this limitation?
3. Give an example of a collision in which the momentum of the system of the colliding objects is not conserved. Explain why momentum is not conserved here but is conserved in other situations.
4. If a hiker were stranded on a frictionless sheet of ice, could she use her knowledge of momentum to get to the edge of the sheet?
5. Why is follow-through important in the swing of a golf club or a baseball bat?
6. How do active head restraints in vehicles reduce injuries from accidents?
7. For traffic safety, is it better to have a traffic light pole that crumples in place or one that shears off at its base when a vehicle crashes into it? Explain your reasoning.
8. Can we ever make automobiles completely safe for humans?
9. How do seatbelts prevent injuries and save lives in car accidents?
10. Why are seatbelts not mandatory for use in school buses? Answer using your knowledge of momentum and collisions.